MA1521 Calculus for Computing

Lecture Group 2

Lecturer: Prof To Wing Keung

Lesson 15: Tuesday 06 October 2026

(Week 8)

Remark. Q15.1 to Q15.8 are on Sections 14.1, 14.2, 14.3 of Thomas' Calculus.

Announcement: No class on Friday 09 October due to NUS Well Being Day.

Section 14.6

Make up arrangement: Do Prof Chin's exercises, get his solution file, and watch his video for L1 Wednesday.

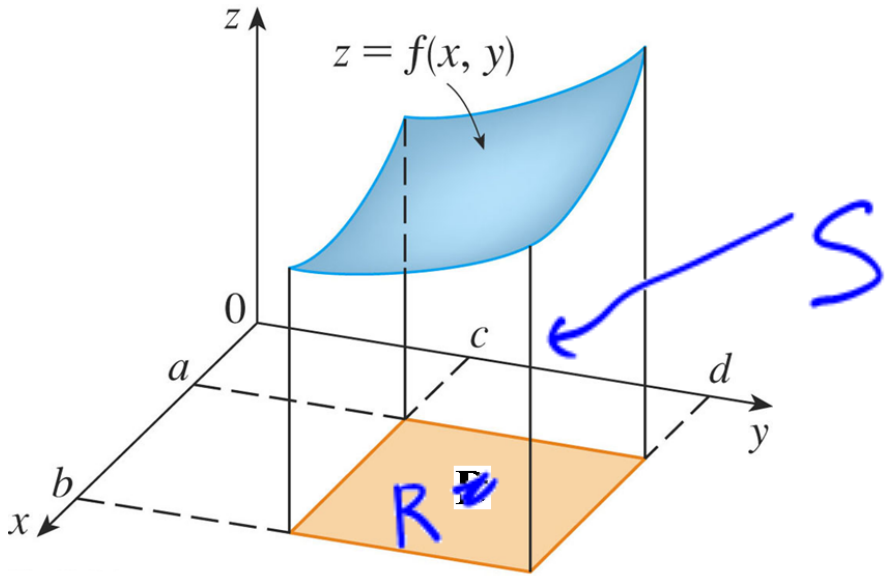
Double Integrals

Suppose $f(x, y)$ is is defined on a closed rectangle

$$R = [a, b] \times [c, d] = \{(x, y) \in \mathbb{R}^2 : a \leq x \leq b, c \leq y \leq d\}.$$

Suppose $f(x, y) \geq 0$. The graph of f is a surface with $z = f(x, y)$ above the region R.

Let S be the solid that lies above R and under the graph of f. How can we find the volume of S?



We mimic the process of defining definite integrals:

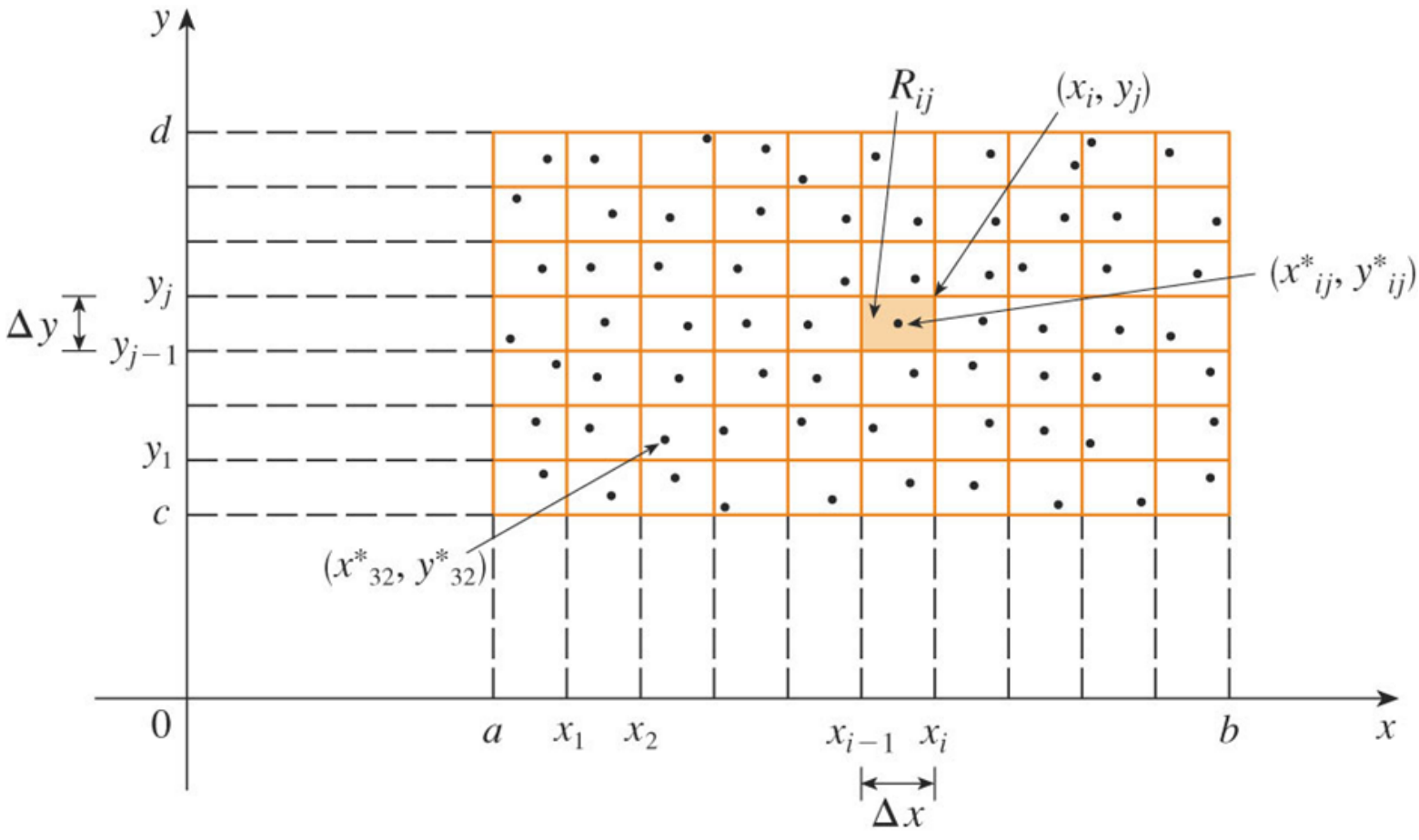
Step 1. Divide R into subrectangles. We do this by

- dividing the interval $[a, b]$ into m subintervals $[x_{i-1}, x_i]$ of equal length $\triangle x = \dfrac{b-a}{m}$, and
- dividing the interval $[c, d]$ into n subintervals $[y_{j-1}, y_j]$ of equal length $\triangle y = \dfrac{d-c}{n}$.

Form subrectangles

$$R_{ij} = [x_{i-1}, x_i] \times [y_{j-1}, y_j], \quad \text{for all } 1 \leq i \leq m, 1 \leq j \leq n.$$

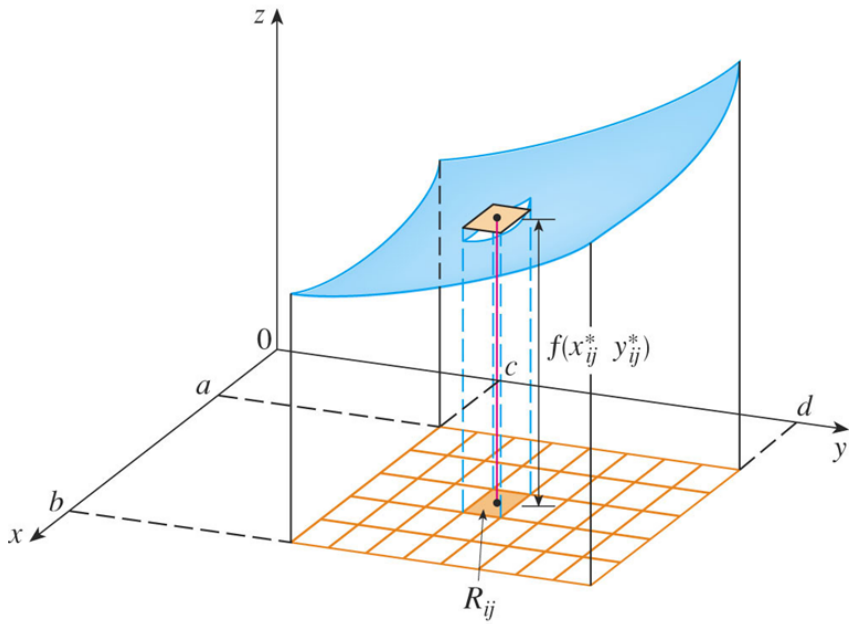
Each of these subrectangles has area $\triangle A = \triangle x \triangle y$.

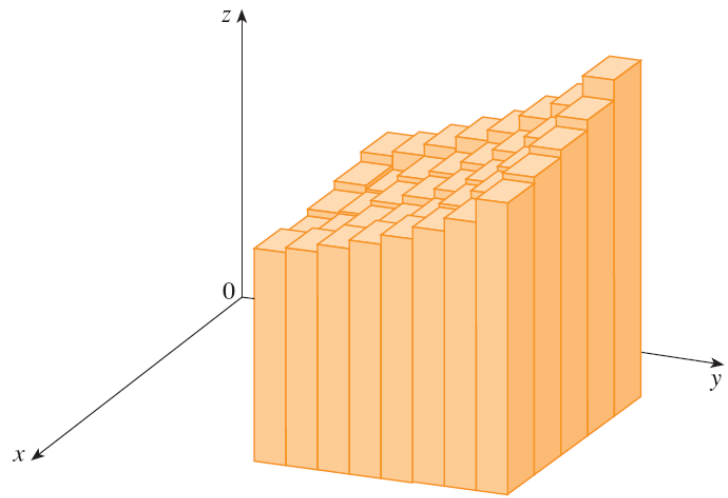

y
d
y_j
Δy
y_{j-1}
y_1
c
0
a
x_1
x_2
x_{i-1}
x_i
b
x
Δx
R_{ij}
(x_i, y_j)
(x^*_{ij}, y^*_{ij})
(x^*_{32}, y^*_{32})

Step 2. Choose a point (x_{ij}^*, y_{ij}^*) in each R_{ij}.

Then approximate the part of S lies above R_{ij} by a thin rectangle box with base R_{ij} and height $f(x_{ij}^*, y_{ij}^*)$.

The volume of this box is given by $f(x_{ij}^*, y_{ij}^*)\triangle A$.



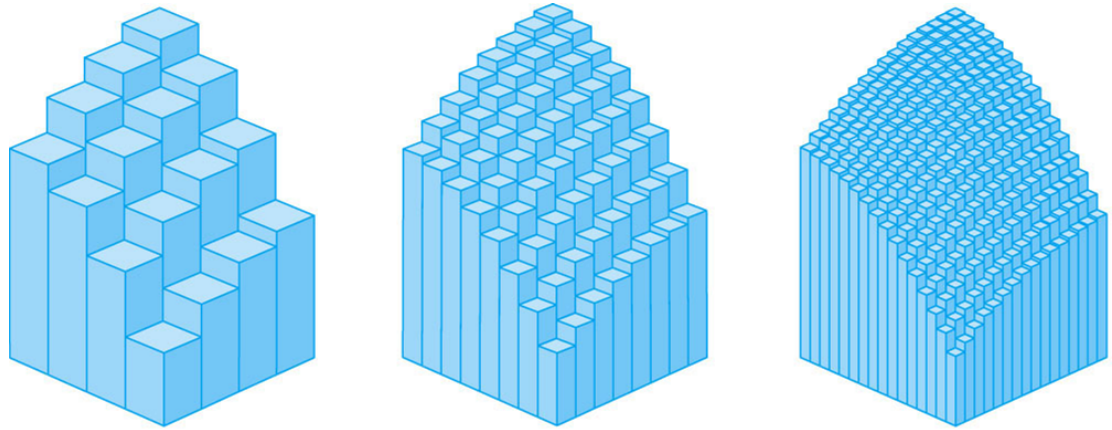


Upon adding (i.e. taking the Riemann sum of) the volumes of all these thin boxes, we get an approximation of the total volume V of S:

$$V \approx \sum_{i=1}^{m} \sum_{j=1}^{n} f(x_{ij}^*, y_{ij}^*) \triangle A.$$

Step 3: For nice functions $f(x, y)$, the approximation becomes better as $m, n \to \infty$. So we get

$$V = \lim_{m,n \to \infty} \sum_{i=1}^{m} \sum_{j=1}^{n} f(x_{ij}^*, y_{ij}^*) \triangle A.$$



(a) $m = n = 4$, $V \approx 41.5$ (b) $m = n = 8$, $V \approx 44.875$ (c) $m = n = 16$, $V \approx 46.46875$

Definition. The **double integral** of f over the region R is

$$\iint_R f(x,y)\,dA = \lim_{m,n\to\infty} \sum_{i=1}^{m}\sum_{j=1}^{n} f(x_{ij}^*, y_{ij}^*)\triangle A.$$

Informally, we think of dA as the area of a small subregion in R, so that $f(x,y)\,dA$ is the volume of the 'thin' solid under the graph of $f(x,y)$ and over the small subregion. Then the double integral sign $\iint_R$ means we take the limit of Riemann sum of the volumes of 'thin' solids.

Summary: If $f(x,y) \geq 0$, the volume V of the solid that lies above the region R and below the surface $z = f(x,y)$ is

$$V = \iint_R f(x,y)\,dA.$$

Fubini's Theorem (First Form).

Suppose $f(x, y)$ is continuous over the rectangular region $R : a \leq x \leq b,\ c \leq y \leq d$. Then

$$\iint_R f(x, y)\, dA \overset{(1)}{=} \int_c^d \left(\int_a^b f(x, y)\, dx \right) dy \overset{(2)}{=} \int_a^b \left(\int_c^d f(x, y)\, dy \right) dx.$$

Informal Justification of (1): For a given y, $CSArea(y) = \int_a^b f(x, y)\, dx$ is the area of a cross section of the solid S under the graph of $z = f(x, y)$ for that y. Then $CSArea(y)\, dy$ is an approximate volume of a 'thin' cross sectional solid with thickness dy. Thus $\int_c^d CSArea(y)\, dy = \int_c^d \int_a^b f(x, y)\, dx\, dy$ is also the volume of the solid under the graph $z = f(x, y)$ over R.

The integrals

$$\int_c^d \left(\int_a^b f(x, y)\, dx \right) dy, \qquad \int_a^b \left(\int_c^d f(x, y)\, dy \right) dx$$

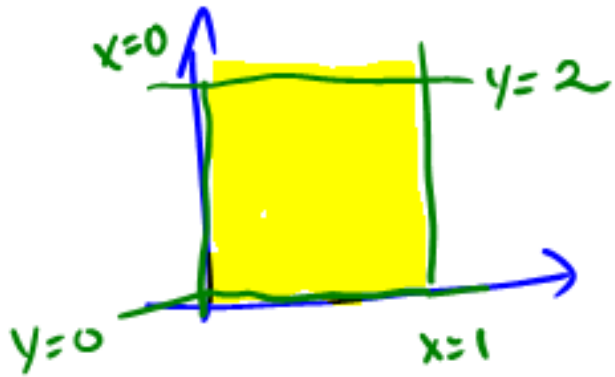
are called **iterated integrals.**

For both type of iterated integrals, we first compute the inner integrals (w.r.t. inner variable), getting an expression in the outer variable.

Then we proceed to evaluate the outer integral, getting finally a number as the answer.

$f(x,y) = x^4 + y^4$

Q15.1. Evaluate the double integral $\iint_R (x^4 + y^4)\, dA$, where R is the region in the xy-plane bounded by the lines $x = 0$, $x = 1$, $y = 0$ and $y = 2$.



$$\iint_R (x^4 + y^4)\, dA = \int_{y=0}^{y=2} \int_{x=0}^{x=1} (x^4 + y^4)\, dx\, dy$$

$$R = \{(x,y) : 0 \le x \le 1,\ 0 \le y \le 2\}$$

Solution: By Fubini's Theorem, we have

$$\iint_R (x^4 + y^4)\, dA$$

$$= \int_0^2 \left(\int_0^1 (x^4 + y^4)\, dx \right) dy$$

$$= \int_0^2 \left[\frac{x^5}{5} + y^4 x \right]_{x=0}^{x=1} dy$$

$$= \int_0^2 \left(\frac{1}{5} + y^4 - (0 + 0) \right) dy$$

$$= \int_0^2 \left(\frac{1}{5} + y^4 \right) dy$$

$$= \left[\frac{y}{5} + \frac{y^5}{5} \right]_{y=0}^{y=2}$$

$$= \frac{2}{5} + \frac{2^5}{5} - (0 + 0) = \frac{34}{5}.$$

$$\iint_R (x^4+y^4)\,dA$$

$$= \int_{x=0}^{x=1}\left(\int_{y=0}^{y=2}(x^4+y^4)\,dy\right)dx$$

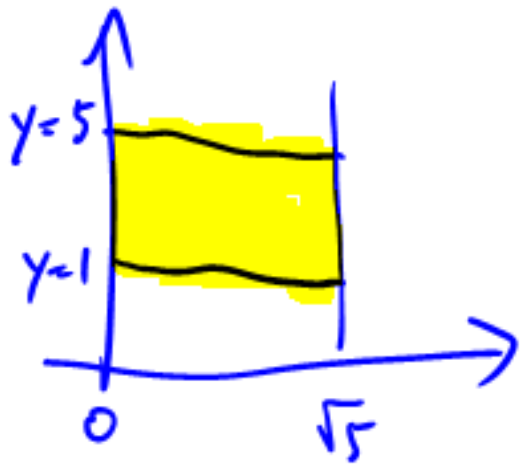
$$= \int_{x=0}^{x=1}\left[x^4y+\frac{y^5}{5}\right]_{y=0}^{y=2}dx$$

$$= \int_{x=0}^{x=1}\left(\left(x^4\cdot 2+\frac{2^5}{5}\right)-(0+0)\right)dx$$

$$= \int_{x=0}^{x=1}\left(2x^4+\frac{32}{5}\right)dx$$

$$= \left[\frac{2x^5}{5}+\frac{32}{5}x\right]_{x=0}^{x=1} = \frac{2}{5}+\frac{32}{5}-(0+0) = \frac{34}{5}.$$

Q15.2. Evaluate the double integral $\iint_R \frac{xy}{\sqrt{9-x^2}}\,dA$, where R is the region in the xy-plane bounded by the lines $x = 0$, $x = \sqrt{5}$, $y = 1$ and $y = 5$.



$$= \iint_R \frac{xy}{\sqrt{9-x^2}}\,dA = \int_{y=1}^{y=5} \left(\int_{x=0}^{x=\sqrt{5}} \frac{xy}{\sqrt{9-x^2}}\,dx \right) dy$$

$$R = \{(x,y) : 0 \le x \le \sqrt{5},\ 1 \le y \le 5\}$$

Solution:

$$\iint_R \frac{xy}{\sqrt{9-x^2}}\,dA$$

$$= \int_1^5 \left(\int_0^{\sqrt{5}} \frac{xy}{\sqrt{9-x^2}}\,dx \right) dy$$

Let $u = 9 - x^2$

$\frac{du}{dx} = -2x$

$x\,dx = \frac{du}{-2}$

$$= \int_1^5 \left[y \cdot \left(-\frac{1}{2}\right) \cdot 2\sqrt{9-x^2} \right]_{x=0}^{x=\sqrt{5}} dy$$

$$= \int_1^5 -y\left(\sqrt{9-5} - \sqrt{9-0}\right) dy$$

$u = 2$ $\quad u = 3$

$$= \int_1^5 y\,dy$$

$$= \left[\frac{1}{2}y^2\right]_{y=1}^{y=5} = \frac{5^2}{2} - \frac{1^2}{2} = 12.$$

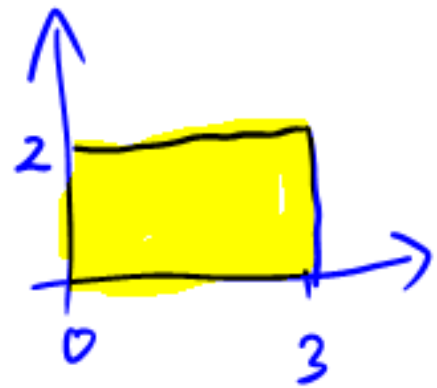
Q15.3. Find the volume of the solid region under the graph of the function $f(x, y) = 3 + (x - 1)^2 + 4y^2$ and above the rectangular region $R : 0 \leq x \leq 3, \ \ 0 \leq y \leq 2$ in the xy-plane.

$R = \{(x,y) : 0 \leq x \leq 3, \ \ 0 \leq y \leq 2\}$

[Remark. Compute the volume as a double integral.]

$$\text{Volume} = \iint_R f(x,y)\, dA$$

$$= \int_{x=0}^{x=3} \left(\int_{y=0}^{y=2} (3 + (x-1)^2 + 4y^2)\, dy \right) dx$$



Solution:

$$\begin{aligned}
\text{Volume} &= \iint_R [3 + (x-1)^2 + 4y^2]\, dA \\
&= \int_0^3 \left(\int_0^2 [3 + (x-1)^2 + 4y^2]\, dy \right) dx \\
&= \int_0^3 \left[3y + (x-1)^2 y + \frac{4}{3} y^3 \right]_{y=0}^{y=2} dx \\
&= \int_0^3 [6 + 2(x-1)^2 + \frac{32}{3}]\, dx = \int_0^3 \left(\frac{50}{3} + 2(x-1)^2 \right) dx \\
&\quad - (0 + 0 + 0) \\
&= \left[\frac{50}{3} x + \frac{2}{3}(x-1)^3 \right]_0^3 \\
&= \frac{50}{3} \cdot 3 + \frac{2}{3}(3-1)^3 - 0 - \frac{2}{3}(0-1)^3 = 56.
\end{aligned}$$

$+ \frac{2}{3}$

Double integral of $f(x, y)$ over general region D

Fubini's Theorem (for Type I region)

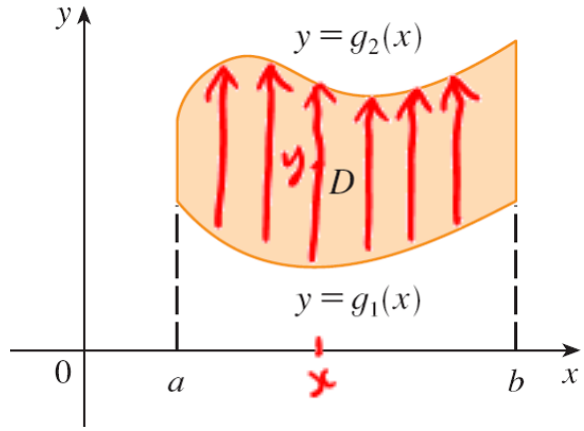
If f is continuous on a Type I region D such that

$$D = \{(x, y) : a \leq x \leq b, \quad g_1(x) \leq y \leq g_2(x)\}$$

y: tricky variable
x: simple variable

then

$$\iint_D f(x, y)\, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\, dy\, dx.$$



Fubini's Theorem (for Type II region)

If f is continuous on a Type II region D such that

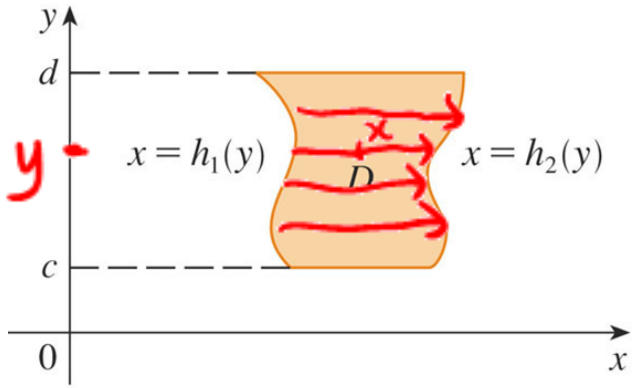
$$D = \{(x, y) : c \leq y \leq d, \ \ h_1(y) \leq x \leq h_2(y)\}$$

then

$$\iint_D f(x, y)\, dA = \int_c^d \left(\int_{h_1(y)}^{h_2(y)} f(x, y)\, dx \right) dy.$$

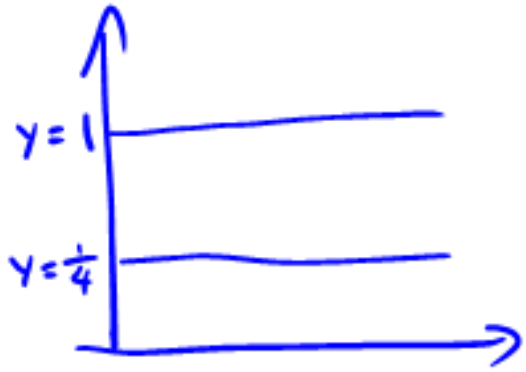
simple variable : y

tricky variable : x



Q15.4. Evaluate the iterated integral

$$\int_{1/4}^{1}\left(\int_{y^4}^{y^2}\sqrt{\frac{y}{x}}\,dx\right)dy.$$



$$= \int_{1/4}^{1}\left[2\sqrt{y}\sqrt{x}\right]_{x=y^4}^{x=y^2} dy$$

$$= \int_{1/4}^{1} 2\sqrt{y}\left(\sqrt{y^2}-\sqrt{y^4}\right) dy$$

$$= \int_{1/4}^{1} \left(2y^{3/2}-2y^{5/2}\right) dy$$

$$= \left[\frac{2y^{5/2}}{5/2}-\frac{2y^{7/2}}{7/2}\right]_{1/4}^{1}$$

Solution:

$$\int_{1/4}^{1}\int_{y^4}^{y^2}\sqrt{\frac{y}{x}}\,dx\,dy = \int_{1/4}^{1}\left(\int_{y^4}^{y^2}\sqrt{\frac{y}{x}}\,dx\right)dy$$

$$= \int_{1/4}^{1}\sqrt{y}\left[2\sqrt{x}\right]_{x=y^4}^{x=y^2}dy$$

$$=2\int_{1/4}^{1}\sqrt{y}(y-y^2)\,dy$$

$$=2\int_{1/4}^{1}(y^{\frac{3}{2}}-y^{\frac{5}{2}})\,dy$$

$$=2\left[\frac{2}{5}y^{\frac{5}{2}}-\frac{2}{7}y^{\frac{7}{2}}\right]_{1/4}^{1}$$

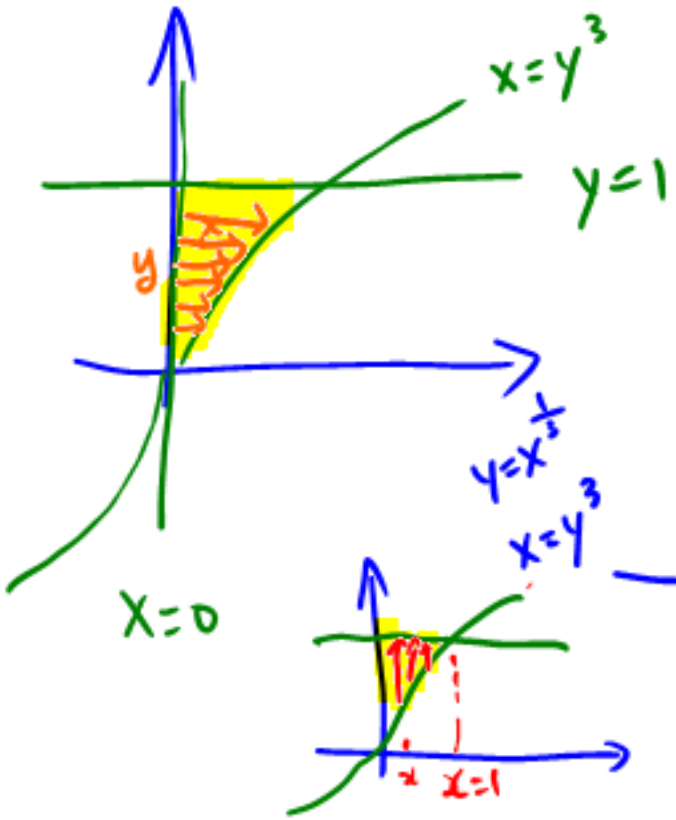
$$=2\left(\frac{2}{5}-\frac{2}{7}\right)-2\left(\frac{2}{5}\cdot\frac{1}{4^{5/2}}-\frac{2}{7}\cdot\frac{1}{4^{7/2}}\right)$$

$$=\frac{233}{1120}.$$

Q15.5. Evaluate the double integral

$$\iint_D y^2 \cos x \, dA,$$

where D is the region bounded by the line $x = 0$, the curve $x = y^3$ and the line $y = 1$.



tricky variable : x

simple variable : y

$$D = \{(x,y) : 0 \le y \le 1,\ 0 \le x \le y^3\}$$

$$\iint_D y^2 \cos x \, dA = \int_{y=0}^{y=1} \int_{x=0}^{x=y^3} y^2 \cos x \, dx \, dy$$

Think of y as tricky variable, x as simple variable

$$D = \{(x,y) : 0 \le x \le 1,\ x^{\frac{1}{3}} \le y \le 1\}$$

$$\iint_D y^2 \cos x \, dA = \int_0^1 \int_{x^{\frac{1}{3}}}^1 y^2 \cos x \, dy \, dx$$

Solution: $D = \{(x, y) : 0 \le y \le 1,\ \ 0 \le x \le y^3\}$.

Then

$$\iint_D y^2 \cos x \, dA$$

$$= \int_{y=0}^{y=1} \left(\int_{x=0}^{x=y^3} y^2 \cos x \, dx \right) dy$$

$$= \int_0^1 \Big[y^2 \sin x \Big]_{x=0}^{x=y^3} dy$$

$$= \int_0^1 (y^2 \sin(y^3) - 0) \, dy$$

$$= \left[-\frac{1}{3} \cos(y^3) \right]_0^1$$

$$= -\frac{1}{3}(\cos(1) - \cos 0) = \frac{1}{3}(1 - \cos(1)).$$

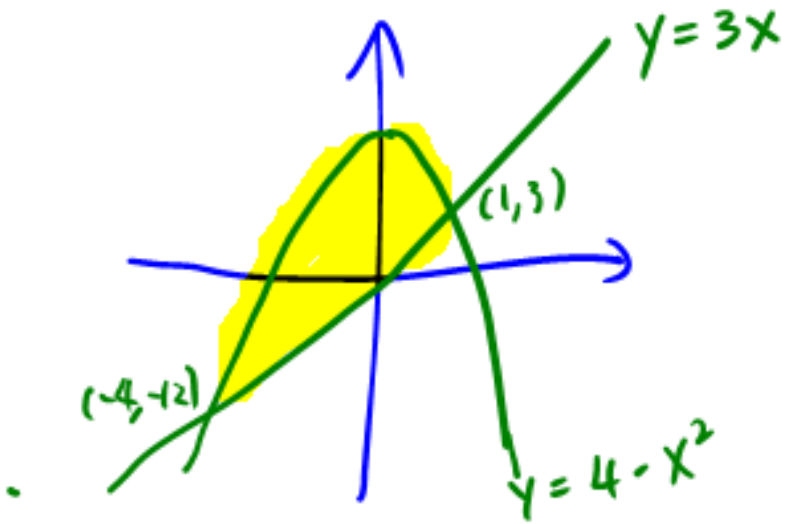
Make substitution $u = y^3$

Q15.6. Evaluate the double integral

$$\iint_D (x + 4)\, dA,$$

where D is the region in the xy-plane that is bounded by the parabola $y = 4 - x^2$ and the line $y = 3x$.

[Remark: You need to find the points of intersection of the parabola and the line.]



Solution:

To describe D, we find the points of intersection of the parabola $y = 4 - x^2$ and the straight line $y = 3x$ by solving their equations simultaneously:
Substituting $y = 3x$ into $y = 4 - x^2$, we get

$$3x = 4 - x^2 \implies x^2 + 3x - 4 - 0$$

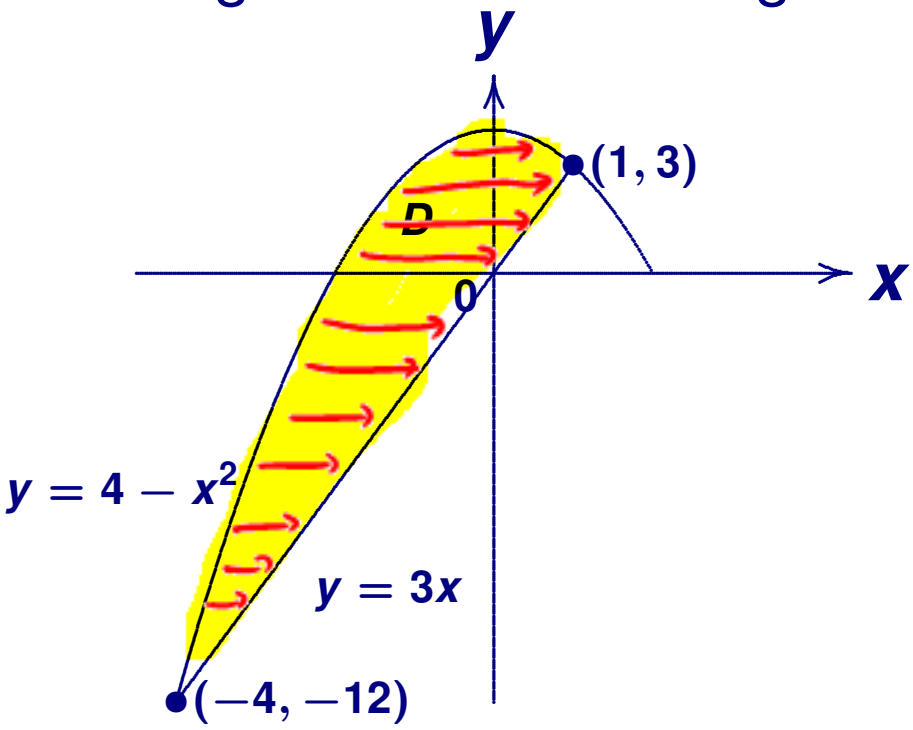
$$\implies (x - 1)(x + 4) = 0 \implies x = 1, -4.$$

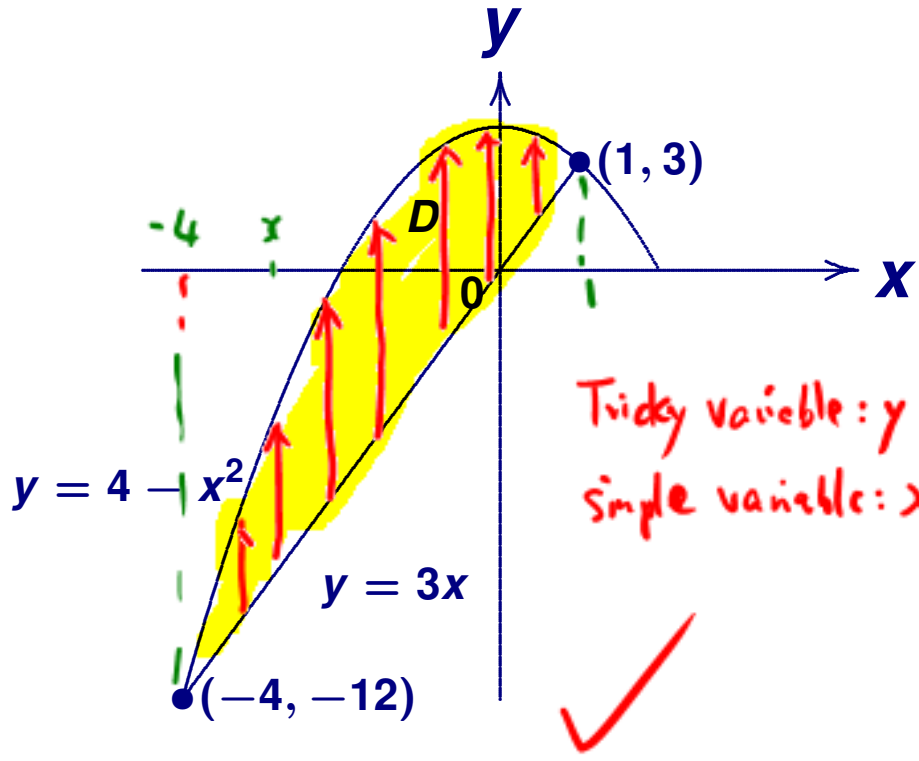
When $x = 1$, $y = 3 \cdot 1 = 3$.

When $x = -4$, $y = -12$.

Thus the points of intersection are $(1, 3)$, $(-4, -12)$.

A rough sketch of the region D is shown below:





$$D = \{(x,y) : -4 \le x \le 1,\ 3x \le y \le 4 - x^2\}$$

$$\iint_D (x+4)\,dA = \int_{x=-4}^{x=1} \int_{y=3x}^{y=4-x^2} (x+4)\,dy\,dx$$

D can be regarded as type I region

$$D = \{(x, y) : -4 \le x \le 1, \quad 3x \le y \le 4 - x^2, \}.$$

We have

$$\iint_D (x + 4)\, dA$$

$$= \int_{-4}^{1} \left(\int_{3x}^{4-x^2} (x + 4) dy \right) dx$$

$$= \int_{-4}^{1} \Big[(x + 4) y \Big]_{3x}^{4-x^2} dx$$

$$= \int_{-4}^{1} (x + 4)(4 - x^2 - 3x) dx$$

$$= \int_{-4}^{1} (-x^3 - 7x^2 - 8x + 16) dx$$

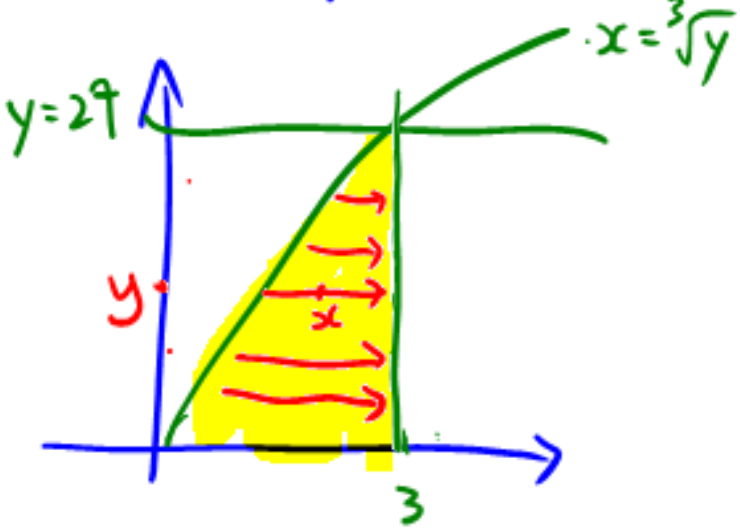
$$= \left[-\frac{1}{4}x^4 - \frac{7}{3}x^3 - 4x^2 + 16x \right]_{-4}^{1} = \frac{625}{12}.$$

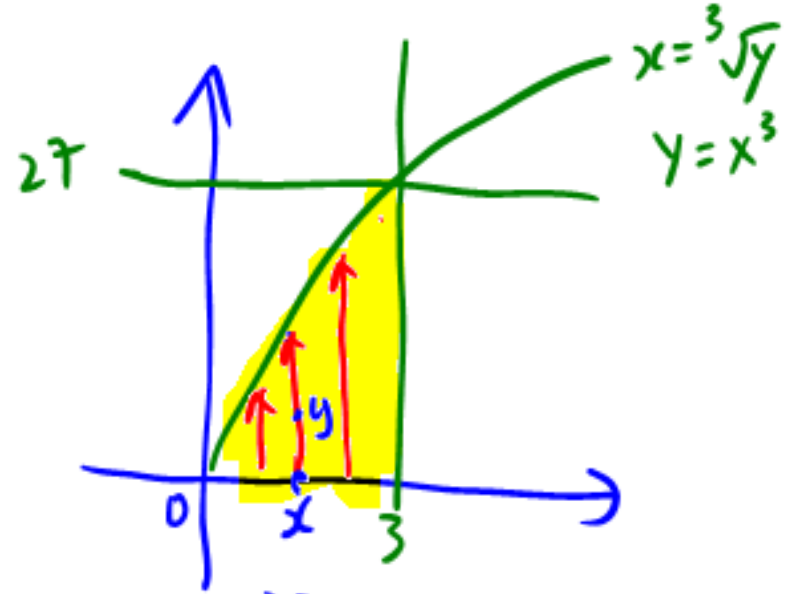
Q15.7. Evaluate the following integral by reversing the order of integration.

$$\int_0^{27}\left(\int_{\sqrt[3]{y}}^{3}\left(\frac{1}{81}+e^{x^4}\right)dx\right)dy. \quad = \int_{x=0}^{x=3}\int_{y=0}^{y=x^3}\left(\frac{1}{81}+e^{x^4}\right)dy\,dx$$

$\int e^{x^4}\,dx = ?$

$D=\{(x,y): 0\le y\le 27,\ \sqrt[3]{y}\le x\le 3\}$





$D=\{(x,y): 0\le x\le 3,\ 0\le y\le x^3\}$

Rewriting R as a type I region given by

$$R = \{(x, y) \in \mathbb{R}^2 \ : \ 0 \leq x \leq 3, \ \ 0 \leq y \leq x^3\}.$$

We have

$$\int_0^{27} \int_{\sqrt[3]{y}}^{2} \left(\frac{1}{81} + e^{x^4}\right) dx\, dy$$

$$= \iint_R \left(\frac{1}{81} + e^{x^4}\right) dA$$

$$= \int_{x=0}^{x=3} \int_{y=0}^{y=x^3} \left(\frac{1}{81} + e^{x^4}\right) dy\, dx$$

$$= \int_0^3 \left(\frac{1}{81} + e^{x^4}\right) \Big[y\Big]_{y=0}^{y=x^3} dx \qquad u = x^4$$

$$= \int_0^3 \left(\frac{x^3}{81} + x^3 e^{x^4}\right) dx = \left[\frac{x^4}{324} + \frac{e^{x^4}}{4}\right]_0^3$$

$$= \left(\frac{81}{324} + \frac{e^{81}}{4}\right) - \left(0 + \frac{1}{4}\right) = \frac{e^{81}}{4}.$$